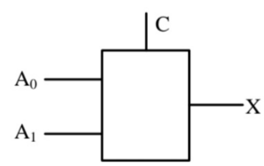
Name: Surabhi P
Id: COMETFWC032
Date: 29 August 2025

`https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/Assembly`

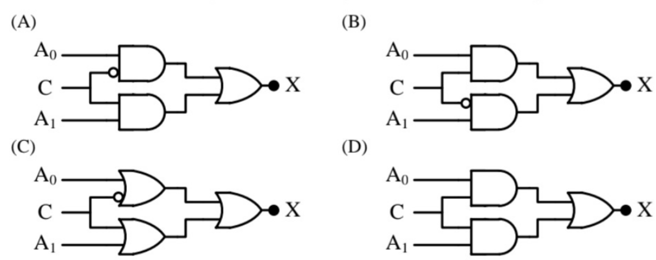
GATE Question solution using assembly

Question

Q.13 In a 2-to-1 multiplexer as shown below, the output X = A_0 if C = 0, and X = A_1 if C = 1.



Which one of the following is the correct implementation of this multiplexer?



Solution

The 2-to-1 multiplexer can be realized using the equation:

$$X = A_1C + A_0\overline{C}$$

This requires two AND gates and one OR gate. One AND gate gets inputs A_1 and C, another gets A_0 and the complement of C, and their outputs are ORed together.

Truth Table

A_1	A_0	C	X
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	1
0	1	1	0
1	0	0	0

AVR Assembly Steps Using Termux

1. Install Termux and AVR tools:

```
pkg update
pkg install proot-distro git curl nano
proot-distro install debian
proot-distro login debian
apt update
apt install avra avrdude
```

2. Write the assembly code implementing $X = A_1C + A_0\overline{C}$, where $A_1 \leftarrow$ PD3, $A_0 \leftarrow$ PD2, $C \leftarrow$ PD4, and $X \rightarrow$ PB5.

3. Assemble to hex:

```
avra mux.asm
```

4. Connect Arduino to Android via USB OTG.

5. Upload hex file:

```
avrdude -p m328p -c arduino -P /dev/ttyACM0 -b 115200 -U flash:w:mux.hex
```

6. Wire respective pins to switch inputs and PB5 to observe output.

Conclusion

The 2-to-1 multiplexer was implemented using the formula $X = A_1C + A_0\overline{C}$ in AVR assembly code. The truth table validates that the logic gates yield the correct output for all possible input combinations. The project demonstrates how digital logic can be programmed, assembled, and executed directly on an Arduino via Termux on Android.